COSC 594/650: Optimization for Machine Learning
Homework 4

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This homework is due on **11/30, 11:59pm**. Format your solution in PDF and submit it on Canvas.

1. (**Logistic regression with quadratic regularization**) Consider the logistic regression optimization problem

$$\underset{w\in\mathbb{R}^d}{\text{minimize}} \quad \frac{1}{n}\sum_{i=1}^{n}\left(\log\left(1+\exp\left(w^T x_i\right)\right)-y_i w^T x_i\right)+\lambda\left\|w\right\|_2^2, \tag{1}$$

where $\lambda > 0$ is a regularization parameter. In this problem, you will use gradient descent to solve (1) and cross validation to tune λ.

   (a) Download the labeled training portion of the public Madelon dataset from the UCI Machine Learning Repository. Use the files `madelon_train.data` and `madelon_train.labels`. This dataset contains 2000 labeled samples and $d = 500$ features. The i-th row of the data matrix is the feature vector x_i. The original labels belong to $\{-1, +1\}$; convert them to $\{0, 1\}$ using

$$y_i \leftarrow \frac{y_i + 1}{2}.$$

   After this conversion, y_i is the label used in (1).
   Solution:

   (b) Using random seed 594, perform a stratified random split of the 2000 samples into a training pool of 1500 samples and a held-out test set of $n_{\text{test}} = 500$ samples. The class proportions should be the same in both sets. From the training pool, construct a stratified training subset $\mathcal{T}_{200}$ containing $n_{\text{tr}} = 200$ samples. Report the number of samples from each class in $\mathcal{T}_{200}$ and in the test set.
   Solution:

(c) Implement gradient descent to solve (1).
Solution:

(d) Choose a grid Λ of positive candidate values for λ. Using random seed 594, perform stratified 5-fold cross validation entirely within $\mathcal{T}_{200}$. For each $\lambda \in \Lambda$, train on four folds and compute the misclassification rate on the remaining fold; repeat until every fold has served once as the validation fold. Select the value of λ with the smallest mean validation misclassification rate. Report Λ, the mean validation rate for each candidate, the selected value of λ, and your rule for breaking any ties. Do not use the held-out test set for any part of this selection procedure.
Solution:

(e) Using the selected value of λ, retrain the classifier on all of $\mathcal{T}_{200}$ and report its misclassification rate on the held-out test set.
Solution:

(f) Construct nested, stratified training sets

$$\mathcal{T}_{200} \subset \mathcal{T}_{700} \subset \mathcal{T}_{1500},$$

where $\mathcal{T}_{700}$ contains 700 samples and $\mathcal{T}_{1500}$ is the complete training pool. Repeat the optimization, cross-validation, and test-evaluation steps independently for $\mathcal{T}_{700}$ and $\mathcal{T}_{1500}$. Use the same candidate grid Λ, random seed, and held-out test set for all three training-set sizes. Tune a separate value of λ for each training set, then retrain on that complete training set before evaluating it on the test set.
Solution:

(g) Based on your observations and our class discussions, describe in a few sentences the relationship between the selected value of λ and the number of training samples n_{tr}. Also describe the relationship between the test error rate and n_{tr}.
Solution:

2. (**Stochastic gradient method**) Consider the optimization problem in (1) using the training set $\mathcal{T}_{1500}$ and the corresponding value of λ selected in the previous problem. At each iteration k, draw an index j_k uniformly at random from $\{1, \ldots, n_{\text{tr}}\}$ and use the stochastic gradient

$$g_k = \left(\frac{\exp\left(w_k^T x_{j_k}\right)}{1 + \exp\left(w_k^T x_{j_k}\right)} - y_{j_k} \right) x_{j_k} + 2\lambda w_k.$$

Let f denote the full objective function in (1), evaluated using all $n_{\text{tr}} = 1500$ training samples. Define

$$e_k \triangleq \log\left(\|\nabla f(w_k)\|_2^2 \right). \tag{2}$$

(a) Implement stochastic gradient descent with a constant step size. Plot e_k versus the iteration number k, and report the constant step size used.
Solution:

(b) Implement stochastic gradient descent with a diminishing step size. Plot e_k versus the iteration number k, and state the diminishing step-size rule used.
Solution: